

# Pranjal Singh

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## EDUCATION

**Indian Institute of Information Technology, Ranchi**

*B.Tech in Electronics & Communication Engineering (CGPA: 8.27)*

2024 – 2028

*Ranchi, Jharkhand*

## WORK EXPERIENCE

**AI Engineer Intern**

Jun 2026 – Present

*Stealthera Innovations Pvt. Ltd.*

*Remote*

- Built and evaluated speech recognition and voice intelligence pipelines using Whisper and ASR frameworks for real-time AI-powered voice solutions.
- Preprocessed voice and audio datasets (cleaning, segmentation, normalization); detected speech patterns, anomalies, and behavioral indicators from raw audio for model training pipelines.
- Fine-tuned AI models and optimized inference performance to improve transcription accuracy and reduce latency in real-time voice use cases.
- Benchmarked multiple model configurations against evaluation datasets, comparing accuracy, speed, and resource usage to select the best-performing setup.
- Documented experiments, model benchmarks, and technical findings, and presented them in weekly engineering reviews to guide the next iteration of the pipeline.

## TECHNICAL PROJECTS

**HELIOS — Enterprise API Engineering Workbench**

2026

*Next.js 14, TypeScript, Turborepo, Ollama (qwen2.5:7b), pnpm Workspaces*

*Live | GitHub*

- Engineered an Enterprise API Workbench compiling OpenAPI 2.0/3.0/3.1 specs into a canonical AST, enabling unified security auditing, multi-language SDK generation, and breaking-change diffing from a single parse.
- Built a privacy-first Local RAG AI Copilot (Ollama, qwen2.5:7b) with an automated Canary Test Suite (`verify:ai`) guaranteeing zero hallucination on API schemas — 100% offline, zero cloud calls.
- Designed a Governance & Compliance Engine auditing APIs against PCI-DSS 4.0, HIPAA, and OWASP Top 10 with 0–100 scoring and auto-generated YAML remediation patches.
- Architected a Turborepo monorepo with 10 decoupled packages sharing one canonical AST and TypeScript types, so the auditor, SDK generator, and diff engine stay consistent and easy to extend.

**ShellMind — Local-First AI CLI for macOS**

2026

*Rust, Zsh, Ollama (qwen2.5:3b), SQLite, macOS Spotlight*

*Live | GitHub*

- Developed a local-first natural language CLI layer for macOS zsh in Rust, translating developer intent into POSIX-safe shell commands with <1ms Tier-1 execution via a deterministic regex engine and Ollama fallback.
- Architected a Type-State Security Boundary (`&ValidatedPlan`) enforcing at compile-time that unvalidated plans cannot reach synthesis; blocked `rm`, `sudo`, and system roots; neutralized shell injection via POSIX single-quoting.
- Implemented Zero-Trust Buffer Injection via `print -z` for mandatory human-in-the-loop inspection; built an Auto-Healing Path Resolver with Levenshtein fuzzy matching; validated with 90 adversarial unit tests.
- Designed a two-tier pipeline where common requests resolve instantly via rules and only complex ones fall back to the local LLM; used Spotlight and SQLite to keep all context on-device.

## TECHNICAL SKILLS

**Languages:** Python, Rust, C++, JavaScript, TypeScript, SQL

**AI / ML:** LLMs, LangChain, Generative AI, NLP, Deep Learning, PyTorch, TensorFlow, Hugging Face, Faster-Whisper, Ollama

**Full-Stack:** Next.js (App Router), React, Node.js, Express, FastAPI, tRPC, TanStack Query, Drizzle ORM, SQLAlchemy, Zod, Tailwind CSS

**Databases & Infra:** PostgreSQL, Neon DB, Supabase, SQLite, Docker, Turborepo, Vercel, Hugging Face Spaces, Git

**Tools:** WebRTC, Web Audio API, MediaPipe, Web Crypto API, Web Workers, REST APIs, OpenAPI/Swagger, Chrome MV3, FFmpeg

## ACHIEVEMENTS

**Smart India Hackathon (SIH):** Qualified the internal rounds to represent IIIT Ranchi at the national level, competing against teams from across the institute.

**Problem Solving:** Solved 200+ DSA problems on LeetCode and CodeChef; active participant in global CodeForces contests, strengthening algorithms and competitive-programming skills.

**CodeChef:** Achieved a maximum rating of **1553** in CodeChef contests.

**Open Source:** Contributed to AOSSIE (Australian Open Source Software Innovation & Education) — resolved bugs and implemented features in full-stack web applications through collaborative code reviews.

**Research:** Experimenting with open-source LLM implementations; focus on LLM architectures, VDFs, and AI automation.